

# Alen Roady (Mujkanovic)

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🌐 [roady.ch](http://roady.ch) (full CV)

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🐙 [github.com/qosmonaut](https://github.com/qosmonaut)

🇨🇭 Swiss/Bosnian dual citizen

🌐 [linkedin.com/in/alenroady](https://linkedin.com/in/alenroady)

📍 Zurich area (open to relocate)



Senior software engineer with 8 years of experience in regulated medtech environments, from start-ups to multinationals. I design, build, and maintain robust software and data platforms aligned with business needs. My goal is to create data-driven solutions that work for the benefit of both patients and healthcare providers.

## Skills

### Tech stack

Expert: Bash | C++ | Python (NumPy, SciPy, PyTorch) | git | GitHub/GitLab | Docker | Balena

Proficient: JavaScript (Node, React) | GCP | PostgreSQL | RabbitMQ | Kubernetes | Django | Qt/PyQt

Intermediate: Rust | C# | Electron | PyTorch | Apache Kafka | Claude Code | Ollama | MCP

### Engineering skills

Expert: scripting | signal processing | embedded | IoT integration | containerization | CI/CD

Proficient: IaC | cloud | DevOps | SW architecture | full stack | databases | machine learning

Intermediate: agentic coding | AI integration | discrete optimization | 3D printing | UI/UX design

### Management skills

Expert: Scrum | technical leadership | stakeholder engagement | Atlassian suite | Adobe CC

Proficient: regulatory (IEC 62304, ISO 13485, IVDR, MDR) | data governance (FHIR, IEC 27701)

Intermediate: strategy | fundraising | market analysis | business growth | manufacturing

### Language skills

Bosnian **native**

English **C2**

German **C2**

French **C1**

Spanish **B2**

## Experience

### Principal SW Engineer – Skaaltec / ETH RELab, Zurich

06.2024 – 01.2026

Founding engineer in an early-stage ETH spin-off developing a wearable neurorehabilitation system for stroke patients. Built and deployed IoT edge devices to partner clinics with Balena Cloud fleet management. Designed and built a user platform and data pipeline. Created a data strategy and ensured regulatory compliance of software products and patient data security. Set up development server, internal tooling, domains and web hosting with continuous deployment using Cloudflare Workers. Supervision of student projects and interns. Left due to chronic lack of funding in a challenging fundraising environment.

C++ | Python | Docker | Rust | Qt | Balena | Django | PostgreSQL | React | Embedded | SW Architecture

### Co-Founder – LoveLabs, Zurich

12.2022 – 09.2023

Co-founded an association to promote expanded access to self-administered rapid diagnostic tests for STIs. Collected market data and performed comparative analysis of commercially available lateral flow devices. Developed go-to-market strategy analyzed product-market fit. Supported fundraising, branding and operations.

Astro | Business Growth | Operations | Regulatory | Fundraising | Partnerships | Web Development

### Software Engineer – Freelance, Remote

09.2021 – 03.2022

Provided technical consulting for the design and development of a medical wearable. Set up an ML training pipeline and provided training to staff SW engineers. Created initial prototype implementation of an embedded ML algorithm for real-time analysis of cardiovascular and kinematic features.

C++ | Python | TinyML | PyTorch | Docker | GCP | Embedded Systems | Technical Training | Scrum

## Software Engineer – Roche Diagnostics [↗](#) (via Cure-IT [↗](#)), Rotkreuz 06.2019 – 06.2021

Designed a digital twin solution and signal processing pipeline for predictive maintenance on the Cobas 6800/8800 PCR system. Developed and compared performance of multiple end-to-end prototypes in pre-development, including IoT edge devices and cloud-based analysis and monitoring applications. Engaged stakeholders and mediated client, business and engineering requirements for product development.

C++ | Python | GCP | Docker | Tensorflow | RabbitMQ | Gitlab CI/CD | Predictive Analytics | Scrum

## Software Developer – Skope Magnetic Resonance Technologies [↗](#), Zurich 07.2015 – 05.2019

Supported initial development of novel magnetic field monitoring technology for MRI applications in an ETH spin-off. Expanded RT signal processing pipeline and visualization of field sensor data. Implemented control API and accompanying SDK for user application development. Set up DevOps process for acquisition application.

LabVIEW | MATLAB | C++ | C# | Jenkins | RT Signal Processing | Data Visualisation | User Support

## Graduate Student Researcher – Harvard Medical School [↗](#), Boston 04.2018 – 10.2018

Developed an optimization framework to generate asymmetric diffusion encoding gradient waveforms for high-resolution cardiovascular DTI. Implemented proof-of-concept on a Siemens Connectome scanner.

Python | MATLAB | CPLEX | Convex Optimization | Pulse Sequence Design | DTI | In Vivo Imaging

## Certificates

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|----------------------------------------------------------------------------------------|-------------|
| Scrum.org: <b>Professional Scrum Product Backlog Management</b> <a href="#">↗</a>      | in progress |
| Scrum.org: <b>Professional Scrum Product Owner I</b> <a href="#">↗</a>                 | in progress |
| Scrum.org: <b>Professional Scrum Master I</b> <a href="#">↗</a>                        | 04.2026     |
| Innosuisse Entrepreneurship Training: <b>Business Growth Medtech</b> <a href="#">↗</a> | 11.2024     |

## Education [↗](#)

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Studies with part-time employment (15h/week) in technical and teaching roles

## MSc Electrical Engineering & Information Technology [↗](#) – ETH Zurich [↗](#) 09.2015 – 11.2018

Biomedical engineering track

- Research projects:**
- Asymmetric Diffusion Encoding for High-Resolution MR-DTI [↗](#)
  - Compressed Sensing Reconstruction for Free-Breathing MR Elastography [↗](#)
  - Real-time Gradient Impulse Response Function Harvesting [↗](#)
  - EXCITE Zurich [↗](#) – biomedical imaging summer school 2017

## BSc Electrical Engineering & Information Technology [↗](#) – ETH Zurich [↗](#) 09.2011 – 12.2016

Signal processing track

- International experience:**
- Chinese University of Hong Kong [↗](#) – exchange semester 2015/16
  - Infineon Technologies [↗](#) – IAESTE internship 2015
  - Peking University [↗](#) – IARU summer school 2014
  - EPF Lausanne [↗](#) – exchange semester 2013/14

## Projects [↗](#) & Passions [↗](#)

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- Agentic coding with cost-effective prompt routing combining local inference and multi-API tooling
- AI-assisted coding on mobile devices: building, testing and deploying on-the-go
- Rock climbing (multi-pitch, sports climbing) and bouldering
- Endurance sports: swimming, cycling, running, hiking, trail running
- Designed and built a trailer sauna from scratch during the COVID-19 lockdowns
- Initiated and co-created a camp at Nowhere, the largest regional Burning Man event in Europe
- Experienced 9 months full-time van life in Europe and Canada
- Facilitate the adoption of stray cats and dogs from Bosnia to families in Switzerland
- Math/physics tutoring for Gymnasium and ETH entrance exams
- Circular economy initiatives and fighting food waste
- Immersive language learning through music, literature and travelling

**Full CV** [↗](#)